Fingerprints Up Close

01 Is your print a loop, a whorl or an arch?



A loop, a whorl and an arch.

Look closer. Find the three prints. One has ridges that curve back, one has a circle, and one has a small hill in the middle.

A little explanation. A loop has ridges that come in and curve back out the same side. A whorl has circles or a spiral. An arch has ridges that flow across with a bump in the middle. Loops are the most common, and arches are the rarest.

Try this. Rub a pencil hard on paper to make a dark gray patch. Press one fingertip on it. Press clear tape on your fingertip, peel it off, and stick it on white paper. Is your print a loop, a whorl or an arch?

02 What tiny details are inside a print?



Up close, ridges end, split and carry tiny pores.

Look closer. Find a ridge that stops, and a ridge that splits in two. Look for the tiny dots along the top of the ridges.

A little explanation. Sweat pores sit in a row along the top of each ridge. Experts look at minutiae, the tiny places where a ridge ends or splits in two. A split is called a bifurcation. They compare these points, not just the pattern. Very rarely, a change in a gene called SMARCAD1 means a person is born with no fingerprints at all.

Try this. Look at your tape print with a magnifying glass if you have one. Find one place where a ridge ends. Then find one place where a ridge splits in two.

03 Why does a koala have fingerprints?



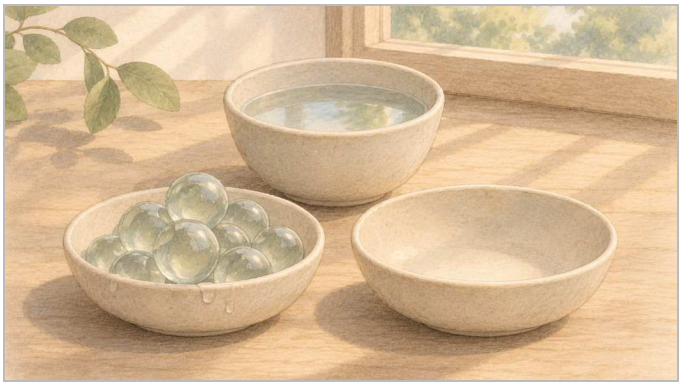
Koalas have prints that look very much like ours.

Look closer. Look at the koala's hand holding the branch. Find the tips of its fingers.

A little explanation. Koala prints look very much like ours. But koalas are only distantly related to us, and their cousins, wombats and kangaroos, do not have prints. Scientists think prints formed separately in both because both grip and handle things.

Try this. Draw a koala hand holding a branch. Then trace your own hand next to it. Tell one way they are alike.

04 Why do fingers get wrinkly in the bath?



Maybe wrinkly fingers help hold wet things.

Look closer. Find the two bowls, the wet marbles, and the bowl of water.

A little explanation. Wrinkly bath fingers are not just skin soaking up water. Your nerves control them. Tiny blood vessels under the skin get narrower, and the skin puckers. Scientists think wrinkles might help us grip wet things, but studies disagree, so for now we say maybe.

Try this. Time how fast you move wet marbles from one bowl to another with dry fingers. Soak your hands in warm water until they wrinkle, then try again. This is a fun try, not proof.